

Streams App – Requirements Document

Version v1.14

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Hierarchy: Stream → Deliverable → Action → Step

Dates: Deliverable = Milestone Date; Action/Step = Due Date

Source Context: Sovereign Cloud streams portfolio PowerPoint (visual style for timeline).

1. Product Vision & Goals

A personal, Monday/Trello-like orchestration app that gives a fast, visual overview of Sovereign Cloud Streams and progress across Deliverables, Actions, and Steps.

Overview-first: a persistent timeline with small balls as the visual anchorpoint at every level; lightweight configuration so you only fill what you need.

Positive date language: 'Milestone Date' for Deliverables; 'Due Date' for Actions and Steps.

Secure access: admin-approved registration and passwordless magic-link MFA.

1.4 Living Spec Policy

This document is living: every decision and rule discussed is integrated and versioned in the Change Log.

2. Key Concepts & Definitions

Stream: Top-level initiative. Stream Milestone Date is computed as the earliest Deliverable Milestone Date.

Deliverable: Tangible output within a Stream. Has Milestone Date.

Action: Work unit under a Deliverable. Has Due Date. Status (Kanban columns): Backlog, To Execute, Executing, Blocked, Delegated, Done, Archive. 'Doing' in summaries maps to Executing.

Step: Micro-step under an Action. Optional Due Date; state Open/Done.

Phases/tags (from PPT): STRATEGY, DESIGN, MODERNIZATION & MIGRATION, MANAGE.

3. User Roles, Access & Registration

Admin (first admin: maarten.bal@capgemini.com): approves registrations; full CRUD; manages roles.

Member: CRUD on Streams/Deliverables/Actions/Steps; self profile management.

Registration creates Pending user; admin approval required.

Passwordless MFA: email magic-link (single-use, 10-minute TTL) required for login.

4. Functional Requirements

4.1 Authentication & Onboarding

Login and Register screens; admin approval flow; magic-link MFA with resend and expiry handling.

4.2 Entities & Hierarchy – Super easy CRUD

CRUD applies to Users, Streams, Deliverables, Actions, Steps.

Parent requirement: Deliverable→Stream; Action→Deliverable; Step→Action.

Minimum viable creation: Name (+ parent for children) is enough. Quick add at every level; inline editing of name/description/status/dates/owners/labels.

Bulk actions: multi-select for status, assignment, labels, archive/delete/restore; optimistic updates and Undo.

Recycle Bin: soft delete by default; restore subtree; hard delete admin-only.

4.3 Overview, Navigation & Context Switching

Main Streams Overview: each Stream shows Deliverable milestone balls on a timeline; Deliverable cards show Doing/Blocked/Delegated counts.

Deep navigation: clicking a Deliverable ball or card goes to Deliverable view; clicking an Action ball or card goes to Action view.

Persistent timeline: The timeline is the persistent visual anchorpoint across all levels; it never disappears during navigation.

Animations: balls 'fall' into the deeper level timeline; respect prefers-reduced-motion.

Breadcrumb header shows path (e.g., Stream 6 ■ Deliverable C ■ Action IV); one-click Up one level; app logo returns to Streams.

Global Actions Kanban mode: all Actions across Streams by status with filters and drag-and-drop.

Global toolbar: Theme (Light/Dark/System), Show descriptions toggle, Command palette (Cmd/Ctrl+K), Search, Filters.

4.4 Progress & Dates (Computation)

Action progress = Steps Done / Steps Total, or derived from status when no Steps (0% Backlog/Blocked/On Hold; 50% In Progress; 100% Done).

Deliverable progress = average of Action progress (or from status if no Actions).

Stream progress = average of Deliverable progress (or from status if no Deliverables).

Stream Milestone Date = min(Deliverable.MilestoneDate); overdue if today > Milestone Date and not Done.

4.5 Timeline & Balls

Level semantics: Stream timeline shows Deliverables at Milestone Dates; Deliverable timeline shows Actions at Due Dates; Action timeline shows Steps at Due Dates.

Default window by level: Stream = 1 year (12 months); Deliverable = 1 year (12 months); Action = 6 months (centered on today by default).

Pan & zoom: users can pan horizontally and zoom (month/quarter/half-year/year); Reset snaps back to default window for the current level.

A Today marker is visible when within range.

No-date shelf: undated children appear as hollow gray balls in a right-side shelf; setting a date animates onto the timeline.

Hover behavior: ball enlarges slightly ($\approx 1.08\times$) and shows a rich tooltip above with full title, full description, date, and status.

Drag-to-reschedule: drag a ball horizontally to change its date (snap to days). Deliverable → Milestone Date; Action/Step → Due Date; Undo available.

Keyboard reschedule: Shift+←/→ (± 1 day), Shift+Alt+←/→ (± 1 week) with live annunciation.

Accessibility: color + shape encoding; keyboard traversal; screen-reader labels.

4.6 Prompt-driven Creation (Builder)

Natural language command bar to create entities and set relationships/dates in one go; parent inference; preview + confirm; Undo.

4.7 Assignments, Notifications & Comments

Owners at any level; comments with @mentions.

Milestone reminders for Deliverables (T■7/T■3/T■1/Day■0/Overdue).

Optional Due Date reminders for Actions/Steps (configurable).

4.8 Search, Filters & Sorting

Global search; filters by status, phase tags, date ranges, owner, overdue/no-date; sorting by Milestone/Due Date, progress, priority, name.

4.9 Import/Export – Excel Mass Import

Template download: Hierarchy (single-sheet) and Multi■sheet templates with an Instructions tab and data validation.

Parent enforcement: children must specify parents (Step → Action + Deliverable + Stream; Action → Deliverable + Stream; Deliverable → Stream).

Dry■run preview: validates rows, shows counts and errors; choose All■or■nothing or Partial import; audit summary and import-level Undo.

4.10 Audit, History & Recycle Bin

Immutable Audit Log (who/what/when; before/after JSON).

Recycle Bin per entity type with restore and hard delete (admin-only).

5. Non■Functional Requirements

Security: HTTPS; magic-link MFA; rate limiting; CSRF/CSP; least privilege; soft delete; audit.

Performance: overview ≤ 2s at 50 Streams/300 Deliverables; CRUD visible update ≤ 150 ms; server confirmation ≤ 500 ms; drag/animations at 60 FPS.

Accessibility: WCAG 2.1 AA; reduced motion respected; keyboard operability everywhere.

Internationalization: en■NL locale; timezone-aware dates; print-friendly light theme for exports.

6. Data Model (Logical)

User: preferences include showDescriptions (bool) and themePreference (light/dark/system).

Stream: fields include computedMilestoneDate, phases[], lastMovementAt, momentumStatus, optional displayKey/ordinal.

Deliverable: streamId FK; milestoneDate; phases[]; owners[]; labels[]; displayKey/ordinal.

Action: deliverableId FK; status enum; dueDate; effort; owners[]; labels[]; kanbanOrder; displayKey/ordinal.

Step: actionId FK; isDone; dueDate; note; owner(optional).

AuditLog: eventType, isMovement, movementScope, parentStreamId, actorId, reason(optional).

7. API (Representative Endpoints)

Auth: POST /auth/register, /auth/approve (Admin), /auth/login, /auth/magic, /auth/logout, /auth/resent-link.

CRUD: standard GET/POST/PATCH/DELETE for Streams/Deliverables/Actions/Steps (soft delete).

Kanban: PATCH /actions/{id}/move (status + kanbanOrder).

Preferences: PATCH /me/preferences (showDescriptions, themePreference).

Import: GET /import/template.xlsx?mode=hierarchy|multisheet; POST /import/dry-run; POST /import/excel; POST /imports/{id}/rollback.

Context: GET /context/{entityType}/{id} (path objects for breadcrumb).

Timeline: entity queries include includeNoDate=true; dates are milestoneDate or dueDate as applicable; level-specific default windows (Stream/Deliverable=1 year; Action=6 months).

Momentum reset: POST /streams/{id}/momentum/reset; POST /streams/{id}/momentum/revert.

8. UI/UX Requirements

Deliverable level: Kanban board (7 columns) + timeline below; Trello-style drag across columns and in-column reorder.

Timeline controls: Zoom (month/quarter/half■year/year), Pan, Today marker, and Reset to default window (Stream/Deliverable=1 year; Action=6 months).

Global Details Toggle: Show/hide descriptions; persists per-user; keyboard shortcut D.

Theme System: Light/Dark/System; WCAG AA colors; print theme always light.

Momentum indicator on Streams Overview: muted (gray■tinted) green/amber/red ball + chip; tooltip shows last movement summary; manual reset.

Keyboard shortcuts: N (quick-add), E (inline edit), Del (soft delete), A (archive), S (status), O (owner), L (labels), D (descriptions), T (theme), Cmd/Ctrl+K (palette), Alt+↑ (up level).

9. Business Rules & Edge Cases

Parent enforcement: children require a parent; context suggests parent automatically.

Soft delete cascade: deleting a parent soft-deletes children; Restore restores subtree.

Protected deletes (optional): prevent delete if item has active children; offer Archive instead.

Duplicate/Clone (optional): cloning a Deliverable optionally clones Actions & Steps with dates cleared.

Momentum thresholds: Active ≤ 7 days; Slowing 8–13; Stalled ≥ 14 ; cosmetic edits (titles/descriptions/labels/owners/dates/reorder/attachments) do not count as movement.

Manual Momentum Reset: sets lastMovementAt=now and momentumStatus=Active; Undo within 10s; rate-limit optional (max 1/day).

10. Notifications & Email Content

Magic-link login email; registration approval email; milestone and due date reminders; optional import summary email.

11. Security Design (Passwordless MFA)

Magic-link tokens: random, hashed, single-use, 10-minute TTL; bound to email + user agent + coarse IP.

Short-lived access token + rotating refresh token with revocation; full audit of admin actions.

12. Tech Stack (Suggested)

Frontend: React (Next.js), TypeScript, Tailwind; Framer Motion/CSS transitions; @dnd-kit for Kanban DnD.

Backend: Node.js (NestJS) or .NET 8; Postgres; Azure App Service; Azure Storage; Azure Key Vault.

Excel parsing: openpyxl/xlsx server-side; batched transactions; preview diff for dry-run.

Accessibility tooling: axe-core checks for both Light/Dark themes.

13. Acceptance Criteria (Summary)

- Create Deliverable shows ball on Stream timeline; Stream Milestone recomputes if earlier.
- Kanban DnD moves Action columns and persists in-column order; Undo works; WIP limit validations optional.
- Timeline hover shows enlarged ball and rich tooltip; drag-to-reschedule updates date with Undo.
- Default timeline windows: Stream=1 year; Deliverable=1 year; Action=6 months; Reset snaps to default; Today marker visible when in range.
- Global details toggle hides descriptions in cards/lists but not in tooltips/panels; persists per-user.
- Theme toggle applies instantly; System mode follows OS; print/PDF always uses light print theme.
- Import dry-run validates and shows errors; partial and all-or-nothing modes; rollback available.
- Momentum: qualifying movements update lastMovementAt and status; cosmetic edits do not; manual reset sets Active and restarts thresholds; Undo reset works.

14. Roadmap

MVP: Auth/approval/MFA; CRUD all; overview+timelines; Deliverable Kanban DnD; progress roll-ups; details toggle; theme; import/dry-run; audit; momentum; reset; timeline defaults.

Phase 2: Prompt builder; PPT-assisted import; ICS exports; advanced reports; per-entity permissions; attachments in import; scheduled imports; role restrictions for momentum reset; WIP limits; remember last zoom per level.

15. Glossary

Milestone Date: positive date for Deliverables; Stream Milestone Date is earliest Deliverable Milestone.

Due Date: planning date for Actions and Steps.

No date shelf: right-of-timeline area showing undated children as hollow balls.

Doing: display label for Executing in roll-ups.

Momentum: Active/Slowing/Stalled indicator based on last qualifying movement.

Appendix A – Seed Data from PPT

Streams and Deliverables mirrored from slides: Digital Sovereignty Assessment (SWM, DSA, DSA&RA;, C4CA), Sovereign Architecture (HCI, SLZ, SSA, EDF, C4CIAr), Sovereign Migrations (SD&CM;, AM&R;, V&AT;, USCE, C4CM), Sovereign Cloud Operations (VLOT, LS&O;, SB&DR;, C4CSO, UCEEM), Sovereign AI Infrastructure (SAIA, SAIF, SAIO), Partnership Management (RPM, LRP, MRC&C;, CRCV, RPDG), Sovereign Cloud Clarity (CVDB, CPV, CF&VP;, CM2T, CCC), Cloud4C Integration (C4CA, C4CIAr, C4CM, C4CSO, C4CAI, C4CSAP).

Appendix B – Visual Conventions

Timeline balls: status colors and shapes (filled, ring for At Risk, hollow for No Date); falling animations; muted momentum colors via gray mixing; accessibility labels; keyboard traversal.